

29. Lexical Analyzer — Identify Tokens

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int isKeyword(char word[])
```

```
{
```

```
    char keywords[10][20] =
```

```
    {
```

```
        "int", "float", "char", "if", "else",
```

```
        "while", "for", "return", "void", "main"
```

```
    };
```

```
    int i;
```

```
    for (i = 0; i < 10; i++)
```

```
    {
```

```
        if (strcmp(word, keywords[i]) == 0)
```

```
            return 1;
```

```
    }
```

```
    return 0;
```

```
}
```

```
int main()
```

```
{
```

```
    char program[] =
```

```
        "int a, b, c = 30;\n"
```

```
"c = a + b;\n";
```

```
char word[50];
```

```
int i = 0;
```

```
int j;
```

```
printf("LEXICAL ANALYZER\n");
```

```
printf("-----\n");
```

```
while (program[i] != '\0')
```

```
{
```

```
    if (isspace(program[i]))
```

```
    {
```

```
        i++;
```

```
    }
```

```
    else if (isalpha(program[i]) || program[i] == '_')
```

```
    {
```

```
        j = 0;
```

```
        while (isalnum(program[i]) || program[i] == '_')
```

```
        {
```

```
            word[j++] = program[i++];
```

```
        }
```

```
        word[j] = '\0';
```

```
        if (isKeyword(word))
```

```

        printf("KEYWORD   : %s\n", word);
    else
        printf("IDENTIFIER : %s\n", word);
}

else if (isdigit(program[i]))
{
    j = 0;

    while (isdigit(program[i]))
        word[j++] = program[i++];

    word[j] = '\0';

    printf("CONSTANT   : %s\n", word);
}

else if (strchr("+-*/%=<>", program[i]) != NULL)
{
    printf("OPERATOR   : %c\n", program[i]);
    i++;
}

else
{
    i++;
}
}

```

```
    return 0;  
}
```

OUTPUT

```
LEXICAL ANALYZER  
-----  
KEYWORD      : int  
IDENTIFIER   : a  
IDENTIFIER   : b  
IDENTIFIER   : c  
OPERATOR     : =  
CONSTANT     : 30  
IDENTIFIER   : c  
OPERATOR     : =  
IDENTIFIER   : a  
OPERATOR     : +  
IDENTIFIER   : b  
  
-----  
Process exited after 1.026 seconds with return value 0  
Press any key to continue . . . |
```